

6-MONTH DAILY STUDY TIMETABLE

AI/ML (Primary Career Skill) + Financial Market Trading (Secondary Skill)

5 October 2026 → 5 April 2027 · 182 days · 26 weeks

Evening Window: 7:00 PM – 11:00 PM · Sustainable college-student workload

CORE PRINCIPLES (Non-negotiable)

- AI/ML is the primary career path → receives 2.5–3 hours every evening.
- Trading is secondary skill/hobby → 1–1.5 hours; never displaces AI/ML or college work.
- Rs 0 real capital during the entire learning period. Paper trading only after Month 3.
- No leverage, no intraday, no F&O, no signal groups, no “guaranteed” strategies.
- Learn → Practice → Build → Review → Test. Theory without code/charts is incomplete.
- Long-term skill compounding over short-term results. Consistency > intensity.

6-MONTH MASTER ROADMAP

Month	Dates	AI/ML Focus (Primary)	Trading Focus (Secondary)	Milestone
1	5 Oct – 4 Nov 2026	NumPy, Pandas, Matplotlib, Statistics, Linear Algebra essentials, Data viz	Market basics, NSE/BSE, NIFTY/SENSEX, Candlesticks, Trends, S/R, Volume	Manipulate datasets; read a chart intelligently
2	5 Nov – 4 Dec 2026	Data preprocessing, Scikit-learn, Supervised learning (Reg + Classif), Model evaluation	Moving averages, RSI, ATR, Basic fundamental analysis, Financial statements	First end-to-end supervised ML pipeline; identify 3 indicators correctly
3	5 Dec 2026 – 4 Jan 2027	Unsupervised learning, Feature engineering, Cross-validation, First ML projects	Risk management, Position sizing, R:R, Psychology, Strategy rules, Journaling	2 complete ML projects; written trading rules + risk plan
4	5 Jan – 4 Feb 2027	Deep Learning fundamentals, PyTorch, Neural nets, CNNs, Training loops	Manual backtesting (50+ setups), Paper-trading journal, Performance stats	CNN image classifier; 50+ backtested setups with stats
5	5 Feb – 6 Mar 2027	Transformers, NLP, Embeddings, RAG, Vector DBs, LLM applications	Python backtesting with Pandas, Walk-forward ideas, Quant concepts intro	Working RAG prototype; coded backtest of one strategy
6	7 Mar – 5 Apr 2027	AI Agents, Deployment (API, Docker), GitHub polish, Substantial final project	F&O; theory only, Algo/quant overview, Full performance review, Edge evaluation	Portfolio-ready final AI app + clear verdict on trading edge

Final Expected Outcome (April 2027)

AI/ML: Strong Python/data foundation · Multiple portfolio projects · DL + LLM/RAG competence · One substantial final application · GitHub portfolio · Deployment basics.

Trading: Solid market fundamentals · Technical + basic fundamental analysis · One defined strategy · 100+ backtested setups · Paper-trading experience · Honest edge evaluation · No real-money losses from learning.

DAILY STRUCTURE & LEARNING METHODOLOGY

Recommended Evening Timetable (Mon–Fri)

Time	Activity	Duration	Focus
7:00 – 8:30 PM	AI/ML Deep Study (Theory + Concepts)	90 min	Books / courses / notes / derivations
8:30 – 8:45 PM	Short Break	15 min	Stand, water, no screens
8:45 – 10:00 PM	AI/ML Practical / Code / Project	75 min	Coding exercises, notebooks, projects
10:00 – 10:15 PM	Short Break	15 min	Reset attention
10:15 – 11:00 PM	Trading Study / Charts / Journal	45 min	Concepts → chart work → journal

Saturday (Project & Review Day – still 4 h max)

7:00–9:00 PM AI/ML project work / revision of weak topics

9:00–9:15 PM Break

9:15–10:15 PM Trading: backtesting + journal + strategy analysis

10:15–11:00 PM Weekly review scorecard + plan next week

Sunday (Lighter but Productive – 2.5–3.5 h)

7:00–8:30 PM AI/ML revision + light project polish

8:30–8:45 PM Break

8:45–10:00 PM Trading journal + chart review + next-week planning

Optional: early finish if energy is low. Consistency beats burnout.

Learning Loop (Apply every week)

AI/ML: Theory → Coding exercise → Small implementation → Mini-project → Review & explain without notes.

Trading: Concept → Mark 8–12 chart examples → Historical analysis → Record hypothetical setups → Journal → Review rules.

Weekly Review Scorecard (Every Sunday)

Area	Questions to answer honestly	Score
AI/ML	What can I code without help? Which concepts are still weak? What did I build this week?	___ / 10
Trading	How many charts analyzed? How many hypothetical trades recorded? Did I follow rules? Mistakes?	___ / 10
Consistency	Days fully completed? Energy management? Any skipped sessions without reason?	___ / 10

Important prioritization rule: If time or energy is short, cut trading first. Never cut AI/ML or college responsibilities. Trading complements technical education; it does not replace it.

Resources Guidance (use free / low-cost first)

AI/ML: Official docs (NumPy, Pandas, Scikit-learn, PyTorch), fast.ai / Hugging Face courses, 3Blue1Brown (linear algebra), StatQuest, Hands-On ML book (Géron), Kaggle Learn, GitHub public repos.

Trading: NSE/BSE investor education, Zerodha Varsity (free), official exchange materials, TradingView free charts, classic books (e.g. “Technical Analysis of the Financial Markets” by Murphy – theory only). Avoid paid signal groups.

MONTH 1 — FOUNDATIONS

Dates: 5 October – 4 November 2026

AI/ML Objective: Build rock-solid Python data skills and the mathematical intuition required for ML.

Trading Objective: Understand what a market is, how the Indian equity market works, and how to read a price chart without guessing.

Key Concepts

AI/ML

- NumPy arrays, broadcasting, vectorization, random, linear algebra ops
- Pandas: Series/DataFrame, indexing, groupby, merge, missing data, time series basics
- Matplotlib & Seaborn: clear, publication-quality plots
- Descriptive statistics, probability distributions, sampling, hypothesis testing intuition
- Linear algebra essentials: vectors, matrices, eigenvalues (conceptual + NumPy)

Trading

- What is a financial market? Equity vs debt vs derivatives (high-level only)
- NSE & BSE structure, indices (NIFTY 50, SENSEX), order types (market, limit, SL)
- Candlestick anatomy, single & multi-candle patterns (observation only)
- Market structure: higher highs/lows, trends, ranges
- Support & resistance: how to mark and why they matter
- Volume: confirmation vs divergence

Weekly Goals

Week	AI/ML Focus	Trading Focus
Week 1 (5–11 Oct)	NumPy deep dive + array projects	Market basics, NSE/BSE, order types
Week 2 (12–18 Oct)	Pandas core + cleaning exercises	Candlesticks + mark 30 examples
Week 3 (19–25 Oct)	Visualization + statistics fundamentals	Trends, S/R, volume on 20 charts
Week 4 (26 Oct–1 Nov)	Linear algebra for ML + mini data project	Integrate chart reading; journal 15 observations
Week 5 (2–4 Nov)	Revision + Month-1 project + checkpoint	Full chart review + self-test

Projects / Exercises

- AI: Clean and explore a public dataset (e.g. Titanic or Iris) end-to-end with NumPy/Pandas/Matplotlib; write a 1-page summary of insights.
- Trading: Create a chart journal (TradingView free or any platform). Annotate 40+ historical charts with trend, S/R, volume notes. No trade signals yet.

Month-End Checkpoint

- AI: Explain broadcasting and groupby without notes; produce 5 clean plots from raw data; solve 10 NumPy/Pandas coding problems.
- Trading: Correctly identify trend direction, mark S/R, and describe volume behaviour on 5 unseen daily charts.

Expected Outcome: I can manipulate real datasets in Python and understand what I am looking at on a stock chart.

MONTH 2 — SUPERVISED LEARNING + INDICATORS

Dates: 5 November – 4 December 2026

AI/ML Objective: Move from data wrangling to building, evaluating and diagnosing supervised models.

Trading Objective: Add quantitative tools (moving averages, RSI, ATR) and begin reading basic financial statements.

Key Concepts

AI/ML

- Train/test split, data leakage, stratification
- Preprocessing: scaling, encoding, imputation pipelines
- Linear & logistic regression, decision trees, random forests, gradient boosting intro
- Metrics: accuracy, precision/recall, F1, ROC-AUC, RMSE/MAE; confusion matrix
- Cross-validation, learning curves, basic hyperparameter search

Trading

- Simple & exponential moving averages – construction and interpretation
- RSI: calculation intuition, overbought/oversold caveats
- ATR: volatility measurement and its use in stop placement (theory)
- Basic fundamental analysis: income statement, balance sheet, cash-flow overview
- Valuation ratios: P/E, P/B, ROE – what they mean and common misuses

Weekly Goals

Week	AI/ML Focus	Trading Focus
Week 6 (5–11 Nov)	Preprocessing pipelines + train/test discipline	Moving averages on 25 charts
Week 7 (12–18 Nov)	Regression project (house prices or similar)	RSI + ATR study + chart examples
Week 8 (19–25 Nov)	Classification project + metrics deep dive	Financial statements walkthrough
Week 9 (26 Nov–2 Dec)	Cross-validation + model comparison	Valuation ratios + 5 company sketches
Week 10 (3–4 Dec)	Revision + Month-2 checkpoint	Indicator integration review

Projects / Exercises

- AI: End-to-end supervised pipeline (regression + classification) with proper train/test, preprocessing, metrics, and a short written diagnosis of model behaviour.
- Trading: Indicator notebook – for 30 historical charts record MA crossover, RSI zone, ATR value and whether price later respected the signal (observation only).

Month-End Checkpoint

- AI: Build a clean pipeline from raw CSV to evaluated model; explain data leakage and why accuracy alone is dangerous.
- Trading: Correctly compute/interpret MA, RSI, ATR on live charts; explain one valuation ratio and its limitations.

Expected Outcome: I can ship a supervised ML pipeline and use three classic indicators without superstition.

MONTH 3 — FEATURE ENGINEERING, PROJECTS & RISK

Dates: 5 December 2026 – 4 January 2027

AI/ML Objective: Master unsupervised techniques, feature craftsmanship and deliver two complete portfolio-style ML projects.

Trading Objective: Internalise risk management, position sizing, psychology and write the first version of personal trading rules.

Key Concepts

AI/ML

- Clustering (K-Means, hierarchical), PCA, dimensionality reduction intuition
- Feature engineering: domain features, interactions, target encoding caveats
- Model selection, pipeline design, scikit-learn best practices
- Project structure: problem statement → data → baseline → iterate → document

Trading

- Risk management: max risk per trade, max daily/weekly loss, correlation of positions
- Position sizing methods (fixed fractional, volatility-based – theory)
- Risk/Reward ratio – why 1:2 or better is preferred; expectancy
- Trading psychology: common biases, journal as the antidote
- Strategy development process: hypothesis → rules → test → review

Weekly Goals

Week	AI/ML Focus	Trading Focus
Week 11 (5–11 Dec)	Unsupervised learning + PCA	Risk rules drafting
Week 12 (12–18 Dec)	Feature engineering deep work	Position sizing exercises (paper)
Week 13 (19–25 Dec)	ML Project 1 (full cycle)	Psychology + journal template
Week 14 (26 Dec–1 Jan)	ML Project 2 + polish	Write first strategy rules document
Week 15 (2–4 Jan)	Month-3 checkpoint & holiday buffer	Rules review + self-assessment

Projects / Exercises

- AI: Two complete projects (e.g. customer segmentation + supervised prediction) with clean notebooks, README, and short blog-style write-up.
- Trading: Personal Risk & Rules document (1–2 pages). Create a structured trading journal template (columns: date, setup, risk, outcome, emotion, lesson).

Month-End Checkpoint

- AI: Present both projects; explain why each feature was engineered; show before/after metrics.
- Trading: Recite risk rules from memory; produce a filled journal page for 10 hypothetical setups; calculate expectancy of a sample set.

Expected Outcome: I have two documented ML projects and a written risk plan I actually intend to follow.

MONTH 4 — DEEP LEARNING + BACKTESTING

Dates: 5 January – 4 February 2027

AI/ML Objective: Transition into neural networks with PyTorch; train and debug simple CNNs.

Trading Objective: Perform systematic manual backtesting of the strategy rules written in Month 3; begin structured paper-trading log.

Key Concepts

- AI/ML**
- PyTorch tensors, autograd, nn.Module, optimizers, loss functions
 - Training loop anatomy, overfitting, regularization (dropout, weight decay)
 - CNNs: convolutions, pooling, classic architectures intuition
 - Transfer learning basics, data augmentation

- Trading**
- Manual backtesting methodology (bar-by-bar, no hindsight)
 - Collecting 50+ historical setups that match written rules
 - Performance statistics: win rate, average R, max drawdown, expectancy
 - Paper-trading process and discipline

Weekly Goals

Week	AI/ML Focus	Trading Focus
Week 16 (5–11 Jan)	PyTorch fundamentals + first nets	Backtest protocol design
Week 17 (12–18 Jan)	CNN from scratch + training	Collect & annotate 25 setups
Week 18 (19–25 Jan)	Transfer learning + regularization	Collect remaining 25+ setups
Week 19 (26 Jan–1 Feb)	CNN project polish + report	Compute performance stats
Week 20 (2–4 Feb)	Month-4 checkpoint	Paper-trading dry run

Projects / Exercises

- AI: Image classification CNN (CIFAR-10 or similar) with training curves, confusion matrix and a short analysis of failure modes.
- Trading: Spreadsheet or notebook of ≥50 backtested setups with entry, stop, target, outcome in R, and summary statistics. Start paper-trading journal.

Month-End Checkpoint

- AI: Train a CNN to reasonable accuracy; explain vanishing gradients and why we use ReLU/BatchNorm.
- Trading: Present the 50-setup stats; answer “Does this set of rules show a positive expectancy on historical data?”

Expected Outcome: I can train a neural network and I possess objective numbers on my first trading strategy.

MONTH 5 — TRANSFORMERS, RAG & QUANT BACKTESTING

Dates: 5 February – 6 March 2027

AI/ML Objective: Understand modern NLP, embeddings, retrieval-augmented generation and build a working RAG prototype.

Trading Objective: Move backtesting into Python; explore basic quantitative ideas; continue paper trading.

Key Concepts

AI/ML

- Tokenization, embeddings, attention mechanism (conceptual + code)
- Transformer architecture high-level, Hugging Face ecosystem
- RAG pipeline: chunking, embedding store, retrieval, generation
- Vector databases (Chroma / FAISS), evaluation of RAG answers

Trading

- Python backtesting with Pandas / vectorbt-style logic (simple)
- Walk-forward thinking and why single backtests overfit
- Basic quantitative concepts: returns distribution, Sharpe intuition, correlation
- Continued paper trading and journal discipline

Weekly Goals

Week	AI/ML Focus	Trading Focus
Week 21 (5–11 Feb)	Embeddings + attention intuition	Python data prep for backtest
Week 22 (12–18 Feb)	Hugging Face models + fine-tune intro	Code first strategy backtest
Week 23 (19–25 Feb)	RAG architecture + vector DB	Walk-forward experiment
Week 24 (26 Feb–4 Mar)	Build RAG prototype end-to-end	Paper-trading review + stats
Week 25 (5–6 Mar)	Month-5 checkpoint	Quant concepts quiz

Projects / Exercises

- AI: RAG application over a personal document collection or public domain texts (e.g. research papers or company filings). Include evaluation of answer quality.
- Trading: Fully coded backtest of the Month-3 strategy on historical daily data; compare results with manual backtest; write a short “limitations” section.

Month-End Checkpoint

- AI: Demo the RAG system; explain why naïve “predict tomorrow’s price with an LLM” is almost always leakage/overfit.
- Trading: Show coded vs manual results; articulate at least three ways a backtest can lie.

Expected Outcome: I have a working RAG system and a coded backtest that I understand the weaknesses of.

MONTH 6 — AGENTS, DEPLOYMENT & FINAL SYNTHESIS

Dates: 7 March – 5 April 2027

AI/ML Objective: Ship a substantial portfolio project that combines multiple skills; learn deployment fundamentals.

Trading Objective: Learn F&O; theory safely, review the entire trading journey, and decide whether a real-money edge exists.

Key Concepts

- AI/ML**
- AI agents / agentic workflows (tool use, planning, memory – conceptual + simple code)
 - FastAPI or similar for serving models
 - Docker basics for reproducible environments
 - Git/GitHub best practices, README, portfolio presentation
 - Cloud deployment fundamentals (high-level)

- Trading**
- Futures & options mechanics (theory only – no trading)
 - Why beginners lose money in F&O; (leverage, time decay, complexity)
 - Full performance review of all paper trades and backtests
 - Honest edge evaluation: does the strategy survive costs and slippage?

Weekly Goals

Week	AI/ML Focus	Trading Focus
Week 26 (7–13 Mar)	Agent concepts + tool-calling	F&O; theory deep dive
Week 27 (14–20 Mar)	API + Docker for ML models	Complete performance review
Week 28 (21–27 Mar)	Final project intensive	Edge evaluation write-up
Week 29 (28 Mar–3 Apr)	Final project polish + GitHub	Trading journey summary
Week 30 (4–5 Apr)	Final checkpoint & celebration	Next 6-month planning

Projects / Exercises

- AI FINAL PROJECT (choose one realistic scope): (A) RAG-powered research assistant with simple agent tools, or (B) End-to-end ML service (train → API → Docker → simple frontend), or (C) Domain-specific classifier + explanation layer. Must include clean repo, README, demo, and short technical write-up.
- Trading: Complete trading journal analysis + 2-page “Edge Report” concluding whether the strategy has a statistical edge after realistic costs.

Month-End Checkpoint

- AI: Live demo of the final project; answer questions on architecture, failure modes and next improvements.
- Trading: Present the Edge Report; state clearly whether real-money trading is justified and under what capital/risk constraints.

Expected Outcome: I possess a portfolio-quality AI application and a clear-eyed assessment of my trading skill.

DETAILED DAILY TASK TEMPLATE & SAMPLE WEEK

Every calendar day follows the same high-level structure, but the concrete task changes. Below is the reusable template followed by a fully specified sample for Week 1. Subsequent weeks follow the monthly/weekly goals listed earlier; you simply replace the specific topic.

Reusable Daily Template (fill in the blanks each evening)

Slot	What to do	Concrete output expected
7:00–8:30 AI Theory	Read / watch / derive today's concept. Take handwritten or typed notes.	1–2 pages of notes or a clear mental model you can explain aloud.
8:45–10:00 AI Code	Implement the concept in a notebook or small script. Solve 2–4 exercises.	Working code cells + short comment on what surprised you.
10:15–11:00 Trading	Study the day's concept → mark examples on charts → journal.	8–12 annotated charts or a filled journal page + one written insight.
Sunday Review	Complete the three scorecards; list next week's priorities.	Scores written + 3 concrete actions for the coming week.

Sample Week 1 (5–11 October 2026) — Fully Specified

Date	Day	AI/ML Session (Theory + Code)	Trading Session	Practical Output
5 Oct	Mon	NumPy: arrays, shape, dtype, indexing, slicing. Code 15 exercises.	What is a market? Equity vs other assets. NSE/BSE overview.	NumPy notebook + 1-page market definition notes
6 Oct	Tue	NumPy broadcasting & vectorization. Speed comparison vs pure Python.	Order types: market, limit, SL, SL-M. Practice on paper.	Broadcasting demo + order-type flashcards
7 Oct	Wed	NumPy linear algebra ops (dot, matmul, inv, eig conceptual).	Candlestick anatomy. Mark open/high/low/close on 15 candles.	Linear-algebra notebook + 15 annotated candles
8 Oct	Thu	Random module, seeds, distributions. Simulate simple random walks.	Single-candle patterns (doji, hammer, engulfing) – observation only.	Random-walk plot + pattern observation sheet
9 Oct	Fri	Mini NumPy project: clean a messy array dataset and compute summary stats.	Multi-candle patterns & basic market structure (HH/HL).	Project notebook + 10 structure charts
10 Oct	Sat	Revision of all NumPy topics + start Pandas Series/DataFrame intro.	Weekly chart review: mark trend on 10 NIFTY daily charts.	Revision notes + 10 trend-marked charts
11 Oct	Sun	Light review + plan Week 2. Scorecard: AI __/10 Trade __/10 Cons __/10	Journal 5 observations from the week. Plan next week's chart work.	Completed scorecard + Week-2 plan

For every subsequent week, replace the specific topics with those listed under the corresponding Month → Weekly Goals. Always keep the same time slots and the Learn → Practice → Build → Review loop. If a college deadline collides, drop the trading hour first.

CRITICAL RULES & ANTI-PATTERNS

Trading – What you will NOT do for 6 months

- No real-money trading of any kind until the Edge Report of Month 6 is positive and capital is risk-capital only.
- No leverage, no margin, no intraday scalping, no F&O; positions.
- No Telegram/Discord signal groups, no “guaranteed” strategies, no copying other people’s trades.
- No increasing position size after a winning streak; no revenge trading after a loss.
- No treating a 50-trade sample as statistical proof; demand larger samples and walk-forward evidence.
- Never risk more than a pre-defined tiny fraction of paper capital on any single idea.

AI/ML – Common traps to avoid

- Do not jump to “predict stock prices with LSTM/Transformer” projects – they almost always contain leakage and unrealistic assumptions.
- Always keep a strict temporal or group split; never randomly shuffle time-series financial data.
- Prefer simple baselines before complex models; a well-tuned logistic regression often beats a poorly regularized neural net.
- Document every experiment; an undocumented notebook is not a portfolio piece.
- Version-control everything with Git from Month 1.

Sustainability for a College Student

Four hours every evening is the maximum. If energy is low, do the AI theory slot and one practical task, then stop. Missing one day is recoverable; missing three consecutive days because of burnout is not. Sleep, exercise and college deadlines always rank above trading. The plan is designed so that even 80 % adherence still produces strong outcomes.

CLOSING NOTE

This timetable is a map, not a prison. The destination is long-term competence in AI/ML and a sober, risk-aware understanding of markets. Six months of deliberate, daily practice compounds into a portfolio and a mindset that most students never develop. Start on 5 October 2026. Track the weekly scores. At the end of April 2027 you will know exactly what you can build and whether trading deserves any of your real capital.

Primary skill → AI/ML career. Secondary skill → trading as a disciplined craft. Capital → Rs 0 until the evidence says otherwise.